

Diagnosis: Get it Right First Time

1. Correct diagnosis? Requires exposure (Cannabis and other substance use, biomass, occupational dusts & fumes etc), symptoms and spirometry

- a) Age 35+, typically 10+ pack year smoking history. Do **CXR & FBC at diagnosis**
- b) Symptoms: progressive breathlessness, cough, sputum production
- c) Obstructive Spirometry: FEV1/FVC < 0.70 or FEV1/VC < 0.70

Nb – airflow obstruction can be missed if the patient does not fully blow out:
Compare FEV1 to both forced and relaxed VC

2. Evidence of primary or co-existent asthma?

Smoking history < 10 pack years, onset before age 35 years, atopy, night-time wakening with cough or wheeze, pronounced wheeze and symptom variability. Substantial variation in FEV1 over time, large response to bronchodilators or oral prednisolone 30mg for 14 days (400+ ml), high FeNO

3. Evidence of primary or co-existent bronchiectasis?

Frequent infective exacerbations, regular mucopurulent sputum, high sputum volume, pseudomonas in sputum culture

When to consider repeating spirometry:

- a) After optimising treatment: is it asthma? Suspect if FEV1 improves by 400 ml+
- b) Following withdrawal of ICS: 3 months or earlier if clinical deterioration
- c) If symptoms substantially worse: **reassess COPD, consider alternative cause**

Record Extended MRC Dyspnoea score, BMI, exacerbation frequency
[Adult-breathlessness-pathway-pre-diagnosis-diagnostic-pathway-support-tool](#)

Key management: COPD Jewels

1. Stop smoking Smoking causes premature death, more frequent flare-ups, worsens existing lung damage, and reduces the benefit of many treatments

2. Pneumococcal, COVID, Annual flu and RSV vaccination (by criteria)

Respiratory vaccination saves lives - the vaccines are dead and cannot cause infection.

3. Pulmonary rehabilitation Consider if MRC breathless scale 2 or more and after hospital admission. Promote physical activity in all

4. Nutrition BMI < 21: advise 3 meals and 3 snacks daily; BMI < 18.5: dietician referral; BMI > 30: advice and referral to local services

5. Check Inhaler technique and concordance see [RightBreathe- Inhalers](#)

6. Infection control see [Reduce COPD flareups](#)

Don't forget:

- a) Patients can complete [ALUK COPD Passport](#) BEFORE their annual review
- b) Patients should have a COPD Self Management Plan
- c) Check oxygen saturations: referral for LTOT if < 92% on 2 occasions (stable for **4 weeks** and sitting at rest for 10 minutes)
- d) Check full blood count when the patient is well to assess eosinophils
- e) Consider co-morbidities: Anxiety (talking therapies, CBT), IHD, AF, heart failure (ECG, BNP/ ECHO), bronchiectasis (HRCT), osteoporosis, CKD (U&E), sleep apnoea & lung cancer (consider CXR & **2WW referral if red flag symptoms**)

COPD flare-ups: Reduce risk: Respiratory vaccines, stop smoking, pulmonary rehab, avoid colds and flu contacts (therapies listed page 2)

Overuse of antibiotics causes resistance: consider antibiotic if substantial increase in sputum volume/ purulence for 2 consecutive days. 1st line Doxycycline or Amoxicillin, 2nd line co-trimoxazole or co-amoxiclav, 3rd line Clarithromycin. 5 days usually sufficient (up to 14 days if co-existent bronchiectasis). Send sputum samples if failed 1st line antibiotic, frequent exacerbations, or concern about bronchiectasis or TB/ NTM (Non-tuberculous mycobacteria).

Overuse of steroids harms patients: consider prednisolone 30mg x 5 days if substantial increase in breathlessness / wheeze for two consecutive days.

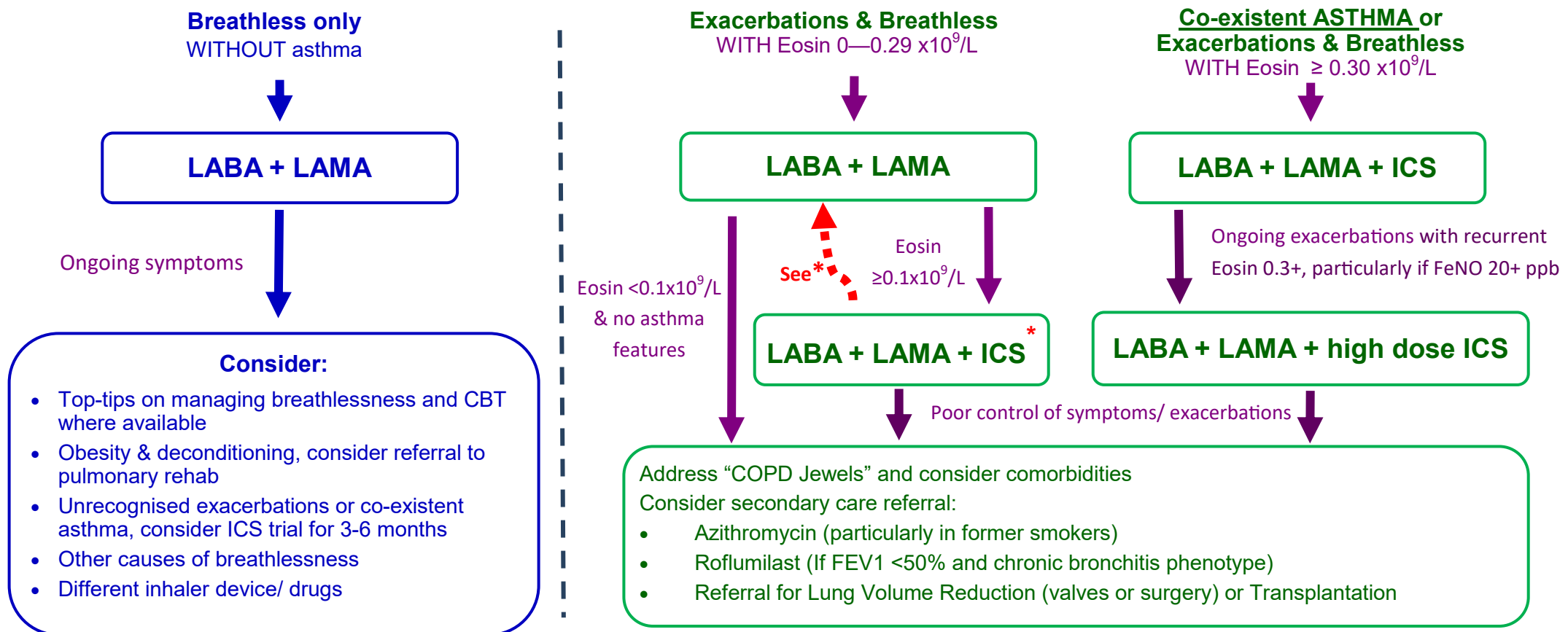
3+ moderate or severe exacerbations in 12 months - consider referral. For every exacerbation provide written and verbal information on exacerbation recognition and management and ensure understood— IF NOT UNDERSTOOD AND THERE ARE CONCERNS OF INAPPROPRIATE USE, DO NOT ISSUE RESCUE PACKS.

COPD Treatable Traits

- ◆ **Breathlessness:** address obesity and deconditioning; consider asthma or other comorbidities (especially cardiac).
- ◆ **Frequent exacerbations** = one hospital admission, or two exacerbations requiring prednisolone and/or antibiotics in the last 12 months.
- ◆ **Eosinophilic COPD:** measure **stable state blood Eosinophil** (Eosin) count (FBC when no acute illness or prednisolone for 4 weeks). If not possible, use the **HIGHEST of at least 3 eosinophil counts within 24 months**. **If Eosin $< 0.1 \times 10^9/L$ and no asthma and no exacerbations avoid ICS.**

COPD Jewels: Pneumococcal, COVID, Annual flu & RSV vaccination (by criteria), Smoking cessation, Pulmonary rehab, Nutrition, Inhaler technique and Concordance

Prescribe SABA for exacerbations / when required rescue:



When prescribing, a single triple inhaler is preferred. If co-existent asthma, a fixed triple may have less ICS than the patient's current ICS+LABA. Check ICS dose

* **Withdrawal of inhaled corticosteroid (ICS)** : Always ensure NO EVIDENCE of co-existent Asthma

- Consider stopping ICS if: **a)** the criteria above are not met or **b)** complications from ICS including recurrent pneumonia with Eosin < 0.3
- It is safe to stop ICS abruptly if the blood eosinophil count has never been 0.3 or higher in the last 12 months

PRESCRIBE all inhalers by BRAND name. Co-existent asthma? CHECK ICS dose when changing to triple.

The most effective inhaler is the one that the patient can and does use: check technique and concordance.

Dry Powder Inhaler (DPI) - lower carbon footprint



Pressurised Metered Dose Inhaler (pMDI) and Soft Mist Inhaler

SABA

Salbutamol Easyhaler®

(Salbutamol 200mcg)

**ONE inhalation
PRN**



Bricanyl Turbohaler®

(Terbutaline 500mcg)

**ONE inhalation
PRN**



Ventolin 200mcg Accuhaler®

(Salbutamol 200mcg)

**ONE inhalation
PRN**



SABA

Salamol Inhaler 100mcg

(Salbutamol 200mcg)

**TWO inhalations
PRN**

Use a **spacer**



Salamol 100 Easi-Breathe® Inhaler

(Salbutamol 200mcg)

**TWO inhalations
PRN**



LABA + LAMA

Ultibro Breezhaler®

(Indacaterol 85mcg /
Glycopyrronium 43mcg)

**ONE inhalation
ONCE daily**



Anoro Ellipta®

(Vilanterol 22mcg /
Umeclidinium 55mcg)

**ONE inhalation
ONCE daily**



Duaklir Genuair®

(Formoterol 12mcg/
Aclidinium 340mcg)

**ONE inhalation
TWICE daily**



LABA + LAMA

Bevespi Aerosphere®

(Glycopyrronium
7.2mcg/
Formoterol 5mcg)

**TWO inhalations
TWICE a day**

Use a **spacer**



Spolto Respimat®

(Tiotropium 2.5mcg/Olodaterol 2.5mcg)

TWO inhalations ONCE daily

Prescribe an Acute prescription for de-
vice & solution cartridge, then 5 repeat
prescriptions for solution only.

Device to be replaced every 6 months.



Triple Therapy LABA + LAMA + ICS

Trelegy Ellipta®

(Fluticasone furoate 92 mcg/
Vilanterol 22 mcg/
Umeclidinium 55 mcg)

**ONE inhalation
ONCE daily**



Trimbow NEXTHaler®

(Beclometasone 88mcg/
Formoterol 5mcg/
Glycopyrronium 9mcg)

**TWO inhalations
TWICE daily**



Triple Therapy LABA + LAMA + ICS

Trixeo Aerosphere®

(Budesonide 160mcg/
Formoterol 5mcg/
Glycopyrronium 7.2 mcg)

TWO inhalations TWICE daily

Use a **spacer**



Trimbow Inhaler 87/5/9

(Beclometasone 87mcg/
Formoterol 5mcg/
Glycopyrronium 9 mcg)

TWO inhalations Twice daily

Use a **spacer**



High Dose ICS + LABA + LAMA (Triple Therapy)

Energair Breezhaler®

(Mometasone 136 mcg/
Indacaterol 114mcg/
Glycopyrronium 46 mcg)

**ONE inhalation
ONCE daily**



Do not use routinely in pure COPD. Consider if persistent exacerbations despite standard triple therapy and COPD Jewels, in patients with: 1) co-existent asthma, OR 2) Eosin ≥ 0.3 , particularly if FeNO 20+ ppb. Specialist review advised.

If no improvement in symptoms or exacerbation frequency, step down to standard dose ICS triple therapy. Caution: higher dose steroids increase the risk of side effects e.g pneumonia.

High Dose ICS + LABA + LAMA (Triple Therapy)

Trimbow Inhaler 172/5/9

(Beclometasone 172mcg/
Formoterol 5mcg/
Glycopyrronium 9 mcg)

TWO inhalations TWICE daily
Use a **spacer**



Do not use routinely in pure COPD. Consider if persistent exacerbations despite standard triple therapy and COPD Jewels, in patients with: 1) co-existent asthma, OR 2) Eosin ≥ 0.3 , particularly if FeNO 20+ ppb. Specialist review advised.

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USEFUL RESOURCES

Techniques to share with patients to help breathlessness

Relax:

- Visualise a positive picture or place
- Distraction – do something to occupy your mind
- Positive self talk – tell yourself this episode will pass

Control your breathing rate:

- Slow down your breathing rate by focusing on long breaths out
- Use abdominal breathing
- Practice Breathing Techniques as listed on the below link:

[Breathing Control](#)

Practice and read the techniques in the above link when you are well so they are familiar to you and you can apply them when you are unwell.



Knowing your patient's baseline status....

Extended MRC Dyspnoea Score

In the past 3 months, when you were **feeling at your best**, which of the following statements best describes your level of breathlessness?

1. Only breathless on strenuous exertion
2. Breathless hurrying or walking up a slight hill
3. Walks slower than contemporaries, or stops after walking on the level for 15 minutes
4. Stops for breath after walking about 100 yards, or for a few minutes on the level
- 5a. Too breathless to leave the house unassisted but independent washing and / or dressing
- 5b. Too breathless to leave the house unassisted and requires help with washing and dressing

Mucolytics and sputum clearance

Refer to NENC At A Glance Consensus Guide, Mucolytics in Adults

Clear sputum using effective coughing and position yourself to help the sputum come up. (i.e. place head and arms on a pillow over a table to expand the chest)

Practice while playing the [Sputum Clearance](#)



Reduce COPD Flare Ups

Simple measures to prevent spread of respiratory viruses

Reduce COPD flare ups and hospital admissions.

See: [Reduce COPD flareups](#)



Asthma and Lung UK:

COPD Passport: [COPD Patient Passport](#)

COPD Self-management Plan:

[Your COPD self-management plan |](#)

[Asthma + Lung UK](#)



Don't forget bones!

Vitamin D for all– recommended OTC.

Additional therapy if confirmed osteoporosis: FRAX score to consider DEXA scan if aged over 50 years with at least one of:

- a) 3 courses prednisolone in one year
- b) Prior fragility fracture
- c) underweight (BMI <18.5 kg/m²)
- d) current smoker, or alcohol > 3u/day

Please access FRAX tool here: [FRAX Tool](#)

Steroid cards

Treatment steroid cards and Emergency Steroid cards are recommended for patients taking doses of inhaled corticosteroids ≥800 micrograms BDP (Beclometasone dipropionate) equivalent per day. All patients taking oral steroids AND inhaled corticosteroids should be provided with a steroid card. Please see CAS Alert link below.

[CAS Alert- Steroid Emergency Card](#)